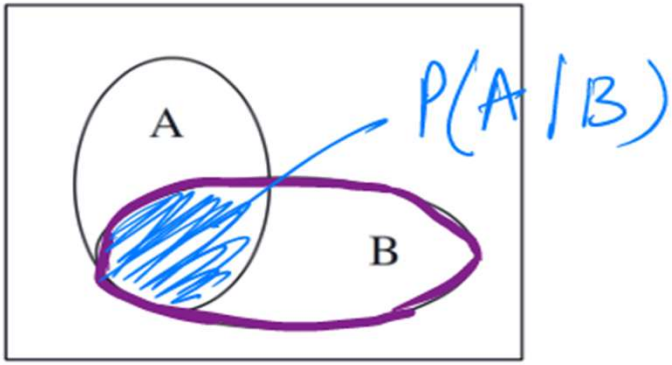
Week 3

Annotations

Conditional Probability: Definition



- $\mathbf{P}(A \mid B)$ = probability of A, given that B occurred
  - B is our new universe
- Definition: Assuming $\mathbf{P}(B) \neq 0$,

$$\mathbf{P}(A \mid B) = \frac{\mathbf{P}(A \cap B)}{\mathbf{P}(B)}$$

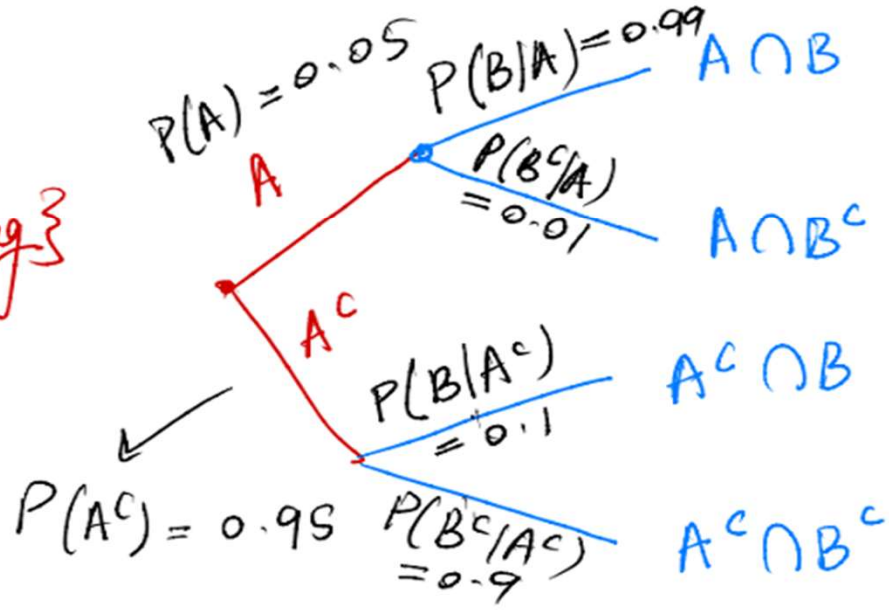
$\mathbf{P}(A \mid B)$ is undefined if $\mathbf{P}(B) = 0$

Example: Flying Under the Radar

Example 1.9. Radar Detection. If an aircraft is present in a certain area, a radar detects it and generates an alarm signal with probability 0.99. If an aircraft is not present. the radar generates a (false) alarm, with probability 0.10. We assume that an aircraft is present with probability 0.05. What is the probability of no aircraft presence and a false alarm? What is the probability of aircraft presence and no detection?

$A = \{\text{Aircraft is present}\}$

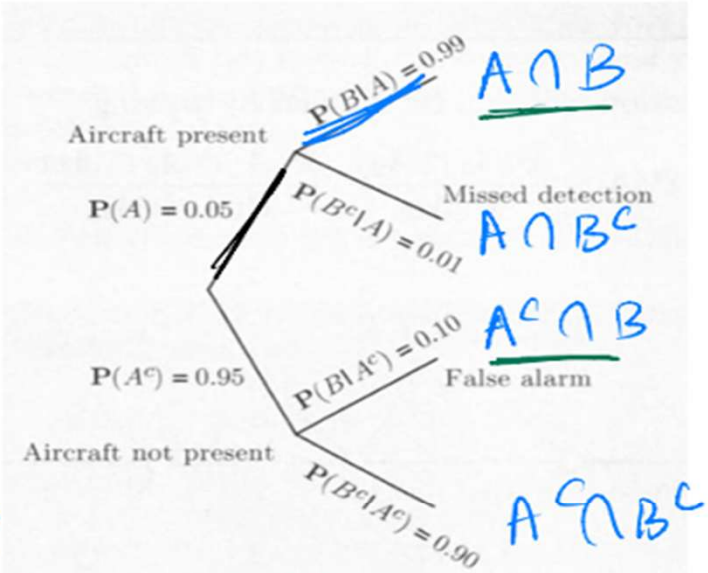
$B = \{\text{Radar detects something}\}$



Example: Flying Under the Radar

Event A: Airplane is flying above

Event B: Something registers on radar screen



Bayes' Rule

Total Prob. Theorem

Multiplication Rule

$P(A \cap B) = P(A) \cdot P(B|A)$

$= 0.05 \times 0.99 = 0.0495$

$P(B) = P(A) \cdot P(B|A) + P(A^c) \cdot P(B|A^c) = 0.0495 + 0.95 \times 0.1$

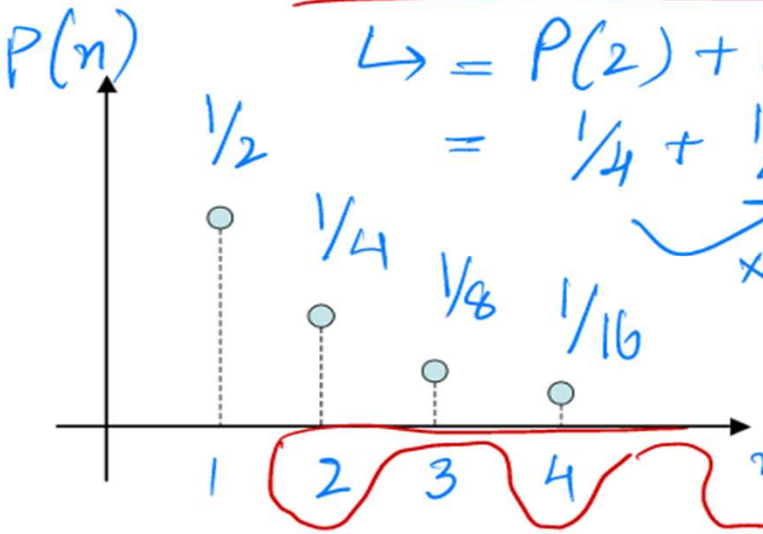
$= 0.1445$

$P(A \mid B) = \dfrac{P(A \cap B)}{P(B)} = \dfrac{0.0495}{0.1445} = 0.34$

Probability Law: Example w/ Countably Infinite Sample Space

- Sample space: $\{1, 2, 3, \ldots\}$
  - We are given $\mathbf{P}(n) = 2^{-n}$, $n = 1, 2, 3, \ldots$
  - Find $\mathbf{P}$(outcome is even)

EXP: "Flip a coin until I get a H. Then I stop"



$\hookrightarrow = P(2) + P(4) + P(6) + \cdots$

$= 1/4 + 1/16 + 1/64 + \cdots$

$\times 1/4$

$= \sum_{k=0}^{\infty} (1/4) \cdot (1/4)^k$

$= \frac{1/4}{1 - 1/4}$

$= 1/3$

Axioms of Probability

1. Nonnegativity: $\mathbf{P}(A) \geq 0$
2. Normalization: $\mathbf{P}(\Omega) = 1$
3. Additivity: If $A \cap B = \emptyset$, then $\mathbf{P}(A \cup B) = \mathbf{P}(A) + \mathbf{P}(B)$

3'. Countable Additivity:

If $A_1, A_2, \ldots$ are disjoint events, then:

$$\mathbf{P}(A_1 \cup A_2 \cup \cdots) = \mathbf{P}(A_1) + \mathbf{P}(A_2) + \cdots$$

↓

These events have to follow a meaningful sequence.

Possibilities for Sample Spaces

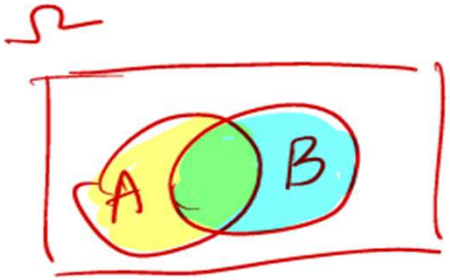
1. Discrete and Finite $\{H, T\}$, $\{1, 2, 3, 4\}$

2. Discrete and Infinite (Countably Infinite)
   - Elements can be arranged in a sequence (i.e. there can be 1-to-1 mapping between elements and integers)
   - Example: Consider repeated coin tosses. After how many coin tosses will you get the first tail.
   - Sample Space = {1, 2, 3, 4, ...}

   → Can apply Countable Additivity Axiom.

3. Continuous (Uncountably Infinite)
   - Elements cannot be arranged in a sequence
   - Example: Throwing a dart on a unit square.

$\{(x, y) \mid 0 \leq x, y \leq 1\}$



$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

A & B Disjoint

$$P(A \cup B) = P(A) + P(B)$$

General Case

$$P(A \cup B) \leq P(A) + P(B)$$

Example 3

Out of the students in a class, 60% love Cricket, 70% love Soccer, and 40% love both Cricket and Soccer. What is the probability that a randomly selected student loves neither Cricket nor Soccer.

1. Exp: "Selecting a Random Student"
2. $\Omega = \{CS, CS^c, C^cS, C^cS^c\}$ ✓✓
3. $P(CS) = 0.4$
4. $P(CS \cup CS^c) = 0.6$
5. $P(CS) + P(CS^c) = 0.6$

$P(CS^c) = 0.6 - 0.4 = 0.2$

Example 3

Out of the students in a class, 60% love Cricket, 70% love Soccer, and 40% love both Cricket and Soccer. What is the probability that a randomly selected student loves neither Cricket nor Soccer.

1. $P(CS \cup C^cS) = P(CS) + P(C^cS)$
2. $0.7 = 0.4 + P(C^cS)$
3. $P(C^cS) = 0.3$
4. $P(\Omega) = P(CS) + P(C^cS) + P(CS^c) + P(C^cS^c)$
5. $1 = 0.4 + 0.3 + 0.2 + P(C^cS^c)$

$P(C^cS^c) = 0.1$